SAMBPS Self-Adaptive Model-Based Protection Systems

PhD Thesis — one-page overview

A hierarchical, physics-aware relay architecture for inverter-rich distribution networks.

Why a New Protection Architecture?

Classical protection relays rest on three assumptions that held for synchronous-generator networks: fault current is *large* (6–10 pu), *passive* (determined by physics, not software), and *directionally predictable* in radial feeders. Inverter-Based Resources (IBRs) violate all three: fast current control caps fault current at 1.1–2.0 pu within 2–5 ms; its magnitude, phase, and sequence composition are set by firmware; and distributed IBRs at every node make feeders bidirectional and meshed. The result is systematic failure of legacy overcurrent, distance, and differential relays — a *protection crisis* worsening as IBR penetration approaches 70–80% of new capacity by 2030. SAMBPS replaces fixed-setting relays with a two-tier adaptive scheme informed by a real-time digital twin.

Two-Tier Architecture

Tier 1 — Local Relay Algorithms *Ch. 4–6*

87L adaptive differential — bias slope $k_m = f(k_{\text{ibr}})$ updated each DT cycle; maintains sensitivity across the full IBR dispatch range.

87T SPRT — sequential-probability-ratio transformer differential; inrush discrimination; IBR-aware adaptive threshold.

Generator suite — elements 40, 64G, 81; IBR mode-detection; confidence-gate veto suppresses spurious trips during fault ride-through.

$\hat{k}_{\text{ibr}}, \hat{\alpha}, \kappa_n$ →

Tier 2 — System Integration *Ch. 7*

5-layer architecture (C16) — digital twin EKF at 1 kHz; estimates k_{ibr}, Z_{line}, fault location α, GFM fraction f_{gfm}; condition-number gate $\kappa_n < 30$.

N-1 coordination (C19) — CTI settings engine adapts relay reach and time-dial to live IBR fleet state via GOOSE broadcast.

HIL validation (C20) — 62-scenario hardware-in-the-loop test suite; IEC 61850 and GOOSE security verified.

Four Problems Addressed

	Problem	What fails / SAMBPS fix
P1	87L threshold failure	Fixed k_m is too conservative at low k_{ibr} (misses high-impedance faults) and too permissive at high k_{ibr} (false restrain during FRT). *Fix:* $k_m = f(k_{\text{ibr}})$ from the 5-layer model, updated every DT cycle.
P2	Distance under-reach	IBR current limiting inflates apparent impedance $Z_{\text{app}} = \alpha Z_L + \frac{I_A+I_B}{I_A} Z_F > \alpha Z_L$; relay under-reaches beyond 60–70% of line. *Fix:* IBR-aware quadrilateral characteristic with online reach correction using $\hat{k}_{\text{ibr}}$.
P3	Microgrid mode transition	50–200 ms gap between grid-connected and islanded modes leaves no valid setting group. *Fix:* mode-triggered setting-group pre-arming synchronised via GOOSE before the transfer.
P4	No real-time IBR visibility	Settings computed offline for a fixed fleet; k_{ibr} varies continuously with irradiance and curtailment. *Fix:* digital twin EKF tracks k_{ibr}, Z_{line}, α at 1 kHz; trusted when $\kappa_n < 30$.

Key Results

- **95.8%** DT–relay agreement (N = 2 000 MC, Study E)
- **0%** FLAG_FP / FLAG_FN in 87L and Z1 regions
- **28.5×** k_{ibr} RMSE gain with phasor-DFT EKF
- **52 ms** fault clearance; DT adds **0 ms** latency
- **62 HIL scenarios** passed; all N-1 contingencies cleared
- $\kappa_n < 30$ gate trusted in **97.3%** of events

Reproduce: `python -m sambp.digital_twin.run_dt_lab -study E -n-mc 2000` | Tag: `sambp-dt-lab-v0.1` | Supersedes TR-43/44/45 standalone scripts.